

Algorithm for file updates in Python

Project description

An algorithm that updates an allow list of which IP addresses get to login in to restricted subnetworks based on who is working with personal patient records.

There is an allow list for IP addresses permitted to sign into the restricted subnetwork.

There's also a remove list that identifies which employees you must remove from this allow list.

Open the file that contains the allow list

```
with open(allow_list, "r") as f:
```

We use the keyword "with" to handle closing the file automatically.

We use the open function with the name of the file and r as arguments and assign the open file to the variable f.

Read the file contents

```
allow_list = "allow_list.txt"
remove_list = ["192.168.205.12", "192.168.6.9"]

with open(allow_list, "r") as f:
    ip_addresses = f.read()
```

Reading the content of the file using the .read() function and storing it in the variable ip_address.

Convert the string into a list

```
ip_addresses_list = ip_addresses.split()
```

Converting using the .split() function.

Iterate through the remove list

```
for i in remove_list:
```

Using a for loop we iterate through the remove list.

Remove IP addresses that are on the remove list

```
for i in remove_list:
    if i in ip_addresses_list:
        ip_addresses_list.remove(i)
```

Using the conditional statement inside the loop to check for IPs in the remove list. In case of finding a positive element it is deleted using the remove function using the same element as an argument. Applying the .remove() method in this way is possible because there are no duplicates in the ip_addresses list.

Update the file with the revised list of IP addresses

```
ip_addresses = "\n".join(ip_addresses_list)

with open(allow_list, "w") as f:
    f.write(ip_addresses)
```

Updating the string by joining the altered list with newline as a separator. Rewriting the entire file using "w" as argument in the open function and the write function using the updated string as an argument.

Summary

In this task I was able to create an algorithm that handles automatic updating of a file. This algorithm would be essential in automating asset management, which is an extremely important aspect of keeping the system secure follows the principle of least privilege and ensuring no leftover privileges cause a HPII breach. The algorithm works as following. First, the file is opened using the with keyword and the open function assigning the file to the f variable. Second, the contents of the f variable are read, turned into a string, and stored inside of the ip_addresses variable. Third, the string variable is turned into a list using the split function and stored into the ip_addresses_list variable. Then the aforementioned list is iterated through using a for loop and each element is checked against the remove list. In case a remove list item was found it is removed using the remove function. Finally, the list is turned back into a string then the allow list is rewritten using the .write() function after opening the file.